MODEL EVALUATION REPORT

Portuguese Bank Marketing Dataset

Task 2 | Predictive modelling and final model selection

Objective. Estimate whether a bank customer is likely to subscribe to a term deposit, supporting more informed marketing prioritisation.

1. Modelling workflow

The notebook mapped the target to a binary outcome, excluded *duration* from pre-call prediction to avoid using information unavailable before the call, performed a stratified train-test split, applied one-hot encoding and scaled numerical variables. Random oversampling was applied to the training set only; the test set remained untouched.

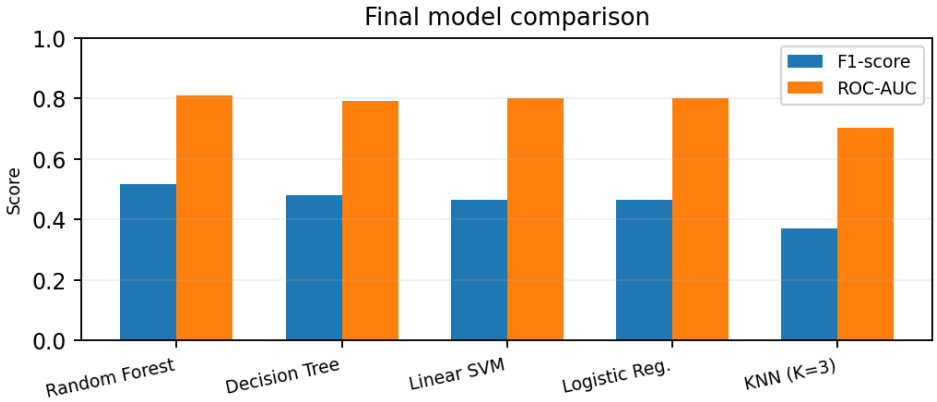
Data component	Rows / dimensions
Training set before oversampling	32,940
Training set after oversampling	58,458 (29,229 per class)
Test set	8,236
Encoded predictors	52

2. Models evaluated

Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN) and Linear Support Vector Machine (SVM) were trained and evaluated on the held-out test set.

3. Final model comparison

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Random Forest	0.8730	0.4521	0.6002	0.5157	0.8086	0.4818
Decision Tree	0.8470	0.3893	0.6293	0.4811	0.7915	0.4332
Linear SVM	0.8311	0.3611	0.6487	0.4640	0.7996	0.4411
Logistic Regression	0.8312	0.3610	0.6466	0.4633	0.7995	0.4457
KNN (K=3)	0.8146	0.3002	0.4849	0.3708	0.7034	0.2349



Selected model: Random Forest. It achieved the highest F1-score (0.5157) and ROC-AUC (0.8086) in the final comparison. Decision Tree and Linear SVM had higher recall, so model choice depends on the bank's balance between finding subscribers and limiting false-positive targeting.

4. Final Random Forest metrics

Metric	Score
Accuracy	87.30%
Precision	45.21%
Recall	60.02%
F1-score	51.57%
ROC-AUC	80.86%
PR-AUC	48.18%
Threshold	0.50

On the 928 positive examples in the test set, the final classification report gave approximately 0.45 precision, 0.60 recall and 0.52 F1-score for the subscription class. This means the model identified about 60% of actual subscribers, while some predicted subscribers did not subscribe.

5. Threshold analysis

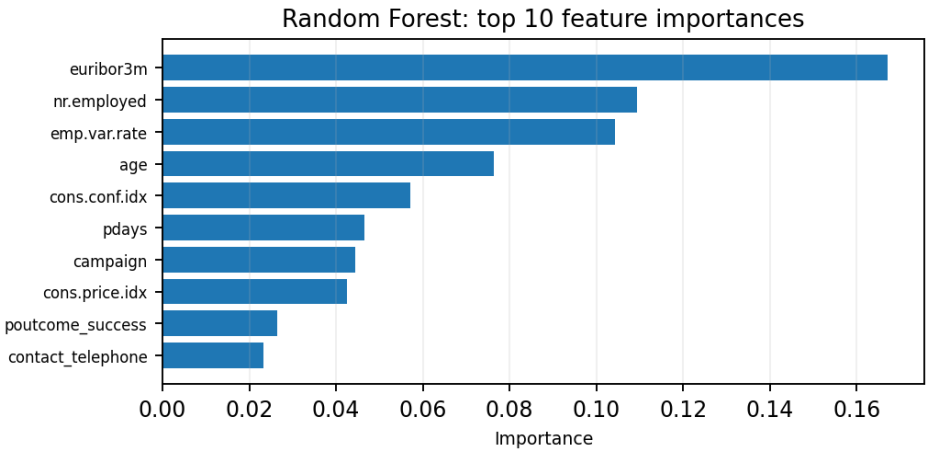
The notebook compared thresholds from 0.2 to 0.7. Among those tested, 0.5 produced the highest F1-score (0.5157). Lower thresholds increased recall but reduced precision; higher thresholds increased precision while reducing recall.

Threshold	Accuracy	Precision	Recall	F1
0.2	0.3859	0.1459	0.9170	0.2518
0.3	0.6582	0.2174	0.7823	0.3403
0.4	0.8309	0.3636	0.6681	0.4709
0.5	0.8730	0.4521	0.6002	0.5157
0.6	0.8865	0.4963	0.5097	0.5029
0.7	0.8947	0.5457	0.3922	0.4564

Threshold selection should reflect campaign costs. A lower threshold may capture more potential subscribers but can increase unnecessary outreach. The 0.5 threshold maximised F1 among the tested values.

6. Feature importance

The Random Forest impurity-based ranking placed *euribor3m*, *nr.employed*, *emp.var.rate*, *age* and *cons.conf.idx* among the leading features.



Permutation importance provided a complementary ranking, led by *euribor3m*, *emp.var.rate*, *nr.employed*, *month_may* and *contact_telephone*. Feature importance describes model reliance, not causation.

7. Model saving and verification

The notebook saved and reloaded *best_random_forest_model.pkl*, *scaler.pkl*, *feature_columns.pkl* and *classification_threshold.pkl*. The verification cell reproduced accuracy 0.8730, F1-score 0.5157 and ROC-AUC 0.8086 at threshold 0.5.

8. Conclusion and deployment note

Random Forest was selected for its strongest F1-score in the final comparison. It can support customer prioritisation but should not be treated as a guarantee of customer behaviour. Monitor campaign outcomes, precision, recall and model drift after deployment.

Deployment status: the model artifacts were saved and verified in the notebook. A hosted web application or production deployment is not evidenced by the notebook alone.